Task 4 — Counterdiabatics: Quantum Annealing for Max-Cut

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I. Challenge overview

In this challenge, we design quantum circuits to solve the Max-Cut problem. The Max-Cut problem is an NP-hard combinatorial optimization problem that seeks to partition the vertices of a graph into two disjoint subsets, such that the total weight of edges crossing between the two subsets is maximized. The problem can be mapped to the following Hamiltonian

$$H_p = \sum_{i<j} J_{ij}\sigma_i^z\sigma_j^z. \tag{1}$$

To find the ground state of a given Hamiltonian, one can use the quantum annealing approach. In this study, we simulate a digitized version of quantum annealing using a simple linear annealing schedule $\lambda(t) = t/T$, where T is the total evolution time. As a next step, we introduce a counterdiabatic term into the time-dependent Hamiltonian based on the first-order adiabatic gauge potential $A_1(t)$. The main goal of this study is to determine the parameters of the initial classical problem and the algorithm parameters for which a performance advantage of the counterdiabatic approach can be observed. As a measure of the algorithm's effectiveness, we consider the probability of obtaining the correct solution (the ground state). The ground state of the problem Hamiltonian is determined beforehand using classical brute-force search.

II. Theoretical description: Ion-trapped native model

Since the target quantum computing platform is ion-trap based, the available native gate set consists of single-qubit rotations and global entangling GMS gates. As our implementation is based on Qiskit, we design the algorithm such that it utilizes only GMS gates of the form $\exp\left(-i\sum_{i<j}\sigma_i^x\sigma_j^x\theta_{ij}/2\right)$. To this end, we replace all σ^z operators in the Hamiltonian of Eq. (1) with σ^x. This transformation allows us to express the problem Hamiltonian in terms of interactions that can be naturally implemented using the available GMS gates on the ion-trap platform. The resulting time-dependent Hamiltonian takes the form

$$H(t) = \lambda(t)\sum_{i<j} J_{ij}\sigma_i^x\sigma_j^x + (1-\lambda(t))\sum_i \sigma_i^z - 2\dot{\lambda}(t)\alpha_1(t)\sum_{i<j} J_{ij}\left(\sigma_j^x\sigma_i^y + \sigma_i^x\sigma_j^y\right). \tag{2}$$

Here the latter term is the adiabatic gauge potential and appears for the counterdiabatic approach; function $\alpha_1(t)$ is taken from Ref. [arXiv:2405.01447v2]; the second term correspond to a simple Hamiltonian with a known ground state. The evolution operator for the digitized quantum annealing (without CD-term) can be obtained by applying the first-order Trotter-Suzuki formula

$$\hat{U}(0,T) \approx \prod_{k=0}^{M}\prod_{n=0}^{N}\left[R_n^z\left(2(1-\lambda(t_k))\Delta t\right)\right]\mathrm{GMS}\left(\{2\lambda(t_k)\Delta t J_{ij}\}_{i<j}\right), \tag{3}$$

where Δt is the trotter step (annealing step), $M = T/\Delta t$, $t_k = k\cdot\Delta t$, the second product is taken over all qubits (N is the total number of qubits), $R_n^z(...)$ is the Z-Pauli gate applied to the n-th qubit, the GMS gate is applied to all qubits, has $N(N-1)/2$ arguments and corresponds to the Qiskit GMS-gate.

For the conterdiabaitc term, we applied additional Z-Pauli gates to implement the CD-term evolution with the Qiskit-based GMS-gates: $R^z(\pi/2)\sigma^x R^z(-\pi/2) = \sigma^y$. For a single trotter step, the layer assosiated with the CD-term has the form

$$\exp\left(-i\sum_{i<j} J'_{ij}\left(\sigma_j^x\sigma_i^y + \sigma_i^x\sigma_j^y\right)/2\right) = \left[\prod_{n=1}^{N} R_n^z\left(\frac{\pi}{2}\right)\mathrm{GMS}\left(J'_{nn+1},...,J'_{nN}\right)R_n^z\left(-\frac{\pi}{2}\right)\right]$$
$$\times\left[\prod_{j=2}^{N} R_j^z\left(\frac{\pi}{2}\right)\mathrm{GMS}\left(J'_{1j},...,J'_{j-1j}\right)R_j^z\left(-\frac{\pi}{2}\right)\right], \tag{4}$$

where $J'_{ij} = -4\Delta t\dot{\lambda}(t_k)\alpha_1(t_k)J_{ij}$.

III. Qiskit-based implementation

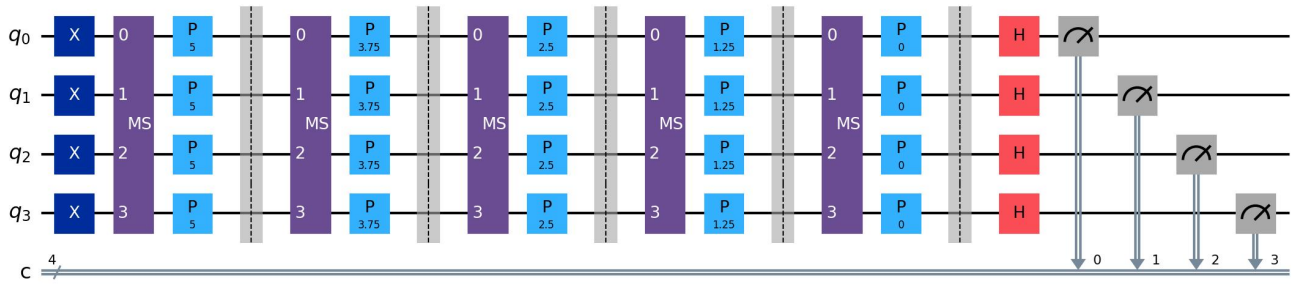

FIG. 1: **Quantum scheme of the digitized quantum annealing.**

Figure 1 shows an exemplary quantum circuit for the digitized quantum annealing algorithm with four qubits. In our implementation, the qubits are initialized in the state $|11...1\rangle$, which is a ground state of the Hamiltonian $\sum_i \sigma_i^z$, see the blue X-gates. Since the target Hamiltonian consists of $\sigma_i^x \sigma_j^x$ interaction terms, the final measurement is performed in the Hadamard basis. The Trotter steps in the circuit are visually separated by the gray regions. As a noise model, we consider a depolarizing channel applied to the GMS gates, while the single-qubit gates are assumed to be noiseless. The quantum circuit corresponding to the case with the counterdiabatic contribution is shown in the Appendix (see Fig. 3).

IV. Results

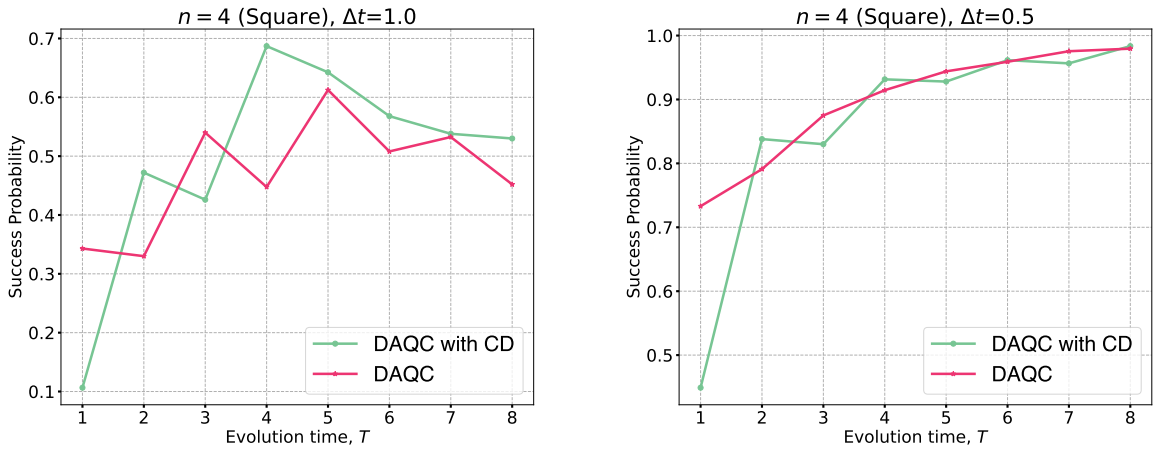

FIG. 2: **Success probability** Dependence of the success probability on the evolution time for standard digitized quantum annealing and for the counterdiabatic approach with different time discretizations ($\Delta t = 1.0$ — left, $\Delta t = 0.5$ — right). The number of qubits (nodes in the graph) is 4.

Figure 2 shows the results obtained from applying the digitized quantum annealing algorithm and the counterdiabatic approach to the Max-Cut problem on a fully connected graph with four nodes. One can observe that the counterdiabatic approach provides an advantage in the final success probability for large time discretization ($\Delta t = 1.0$). In this regime, the standard digitized quantum annealing approach loses its adiabatic properties, and the probabilities of non-ground states increase. When the time discretization is sufficiently small ($\Delta t = 0.5$), the adiabatic properties are preserved, and the counterdiabatic approach does not provide a noticeable advantage for the considered Max-Cut problem.

V. Future research

In the current implementation, we assumed that all qubits can interact with each other. In future work, we plan to consider scenarios that more closely reflect real hardware platforms, where typically only nearest-neighbor interactions are available. In addition, in this study we investigated the influence of the first-order adiabatic gauge potential, while the effects of higher-order terms, such as the second-order contribution, remain an interesting direction for further research.

VI. Appendix

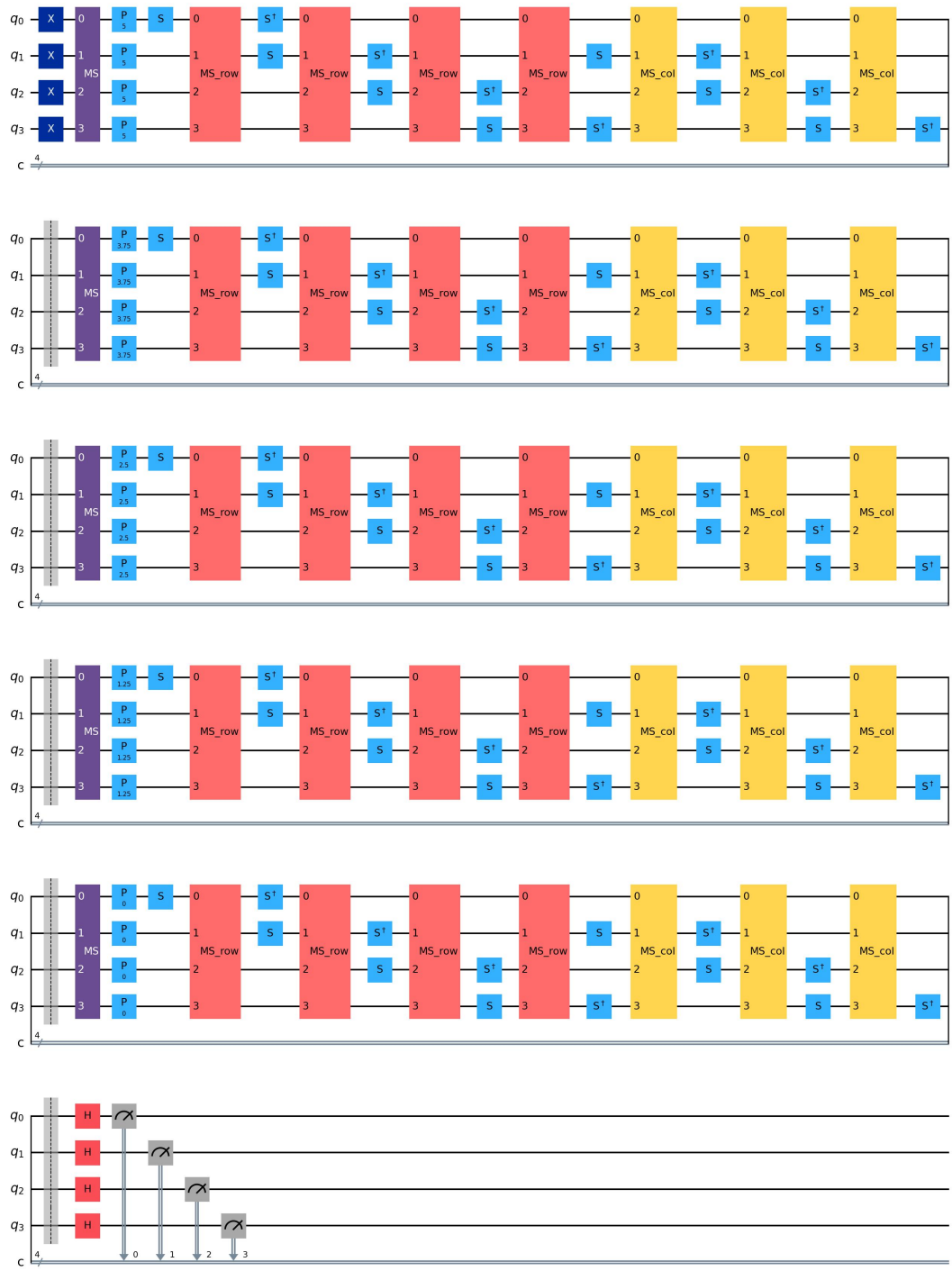

FIG. 3: **Quantum scheme of the digitized quantum annealing with the conterdiabatic term.**